

artifact-2-operational-strategy.md

Operational Strategy

Summary

The operational strategy is built around the principle: every stage must be measurable, retryable, and stoppable before it damages latency, cost, or downstream capacity.

The strategy covers:

- Service health and SLOs.
- Per-stage latency and queue-age monitoring.
- Cost and token usage monitoring.
- OCR, model, and webhook reliability.
- Model quality evaluation.
- Canary release and rollback.
- Incident response and replay.
- Managed Bedrock and future self-hosted GPU inference observability.

Case Study Interpretation and Scope

The case study asks for an operational plan that can prove the system is healthy while it processes large PDFs, handles bursts, controls cost, and supports managed and self-hosted inference.

This artifact covers:

- How we monitor P95 latency, queue age, throughput, cost, model quality, and GPU metrics.
- How we detect and respond to provider throttling, OCR failures, model failures, poison documents, and webhook failures.
- How we run CI/CD, model evaluation, canaries, and rollback.
- How we operate both the v1 managed Bedrock path and the v2 self-hosted GPU path.

Operational Assumptions

Area	Assumption
SLA clock	The 5-minute P95 processing clock starts at <code>accepted_at</code> , after validation and malware scan. Upload time is excluded.
Stage latency budgets	Stage 2 $\leq 30s$, Stage 3 P95 $\leq 180s$, Stage 4 P95 $\leq 60s$, Stage 5 P95 $\leq 30s$, Stage 6 P95 $\leq 10s$, Stage 7 P95 $\leq 10s$. These leave margin within the 5-minute target and drive backpressure decisions.
Traffic	10,000+ docs/day sustained and 50,000-document burst intake.
Tenant model	Metrics, quotas, cost, and operational controls are tenant-scoped
Manual review	Critical failures and degraded outputs may require review before customer delivery. Manual-review backlog is an operational SLO input.
Deployment	All stacks are CDK Python, deployed through GitHub Actions with tests, security checks, and CDK synth validation.

4. Operational Architecture

Operationally, the system has four feedback loops:

Loop	Purpose
Reliability loop	Detect failures, retries, DLQ growth, poison documents, provider outages, and stuck jobs.
Latency loop	Track P50/P95/P99, queue age, slowest required chunk, and end-to-end critical path.
Cost loop	Track OCR pages, model tokens, embeddings, storage, observability, retries, and per-tenant spend.
Quality loop	Track field accuracy, citation validity, hallucination indicators, manual-review rate, and model drift.

5. SLOs and Metrics

5.1 Core SLOs

SLO	Target	Why it matters
Platform availability	99.9%	Case study requirement and customer trust baseline.
End-to-end processing latency	P95 < 5 minutes after accepted_at	Primary latency requirement. Stage budgets: S2 $\leq 30s$, S3 P95 $\leq 180s$, S4 P95 $\leq 60s$, S5 P95 $\leq 30s$, S6/S7 $\leq 10s$ each.
Job success rate	Track by document type and schema	Separates platform failure from manual-review-required documents.
Critical-field quality	No material regression during canary	Critical fields drive customer trust.
Webhook first-attempt latency	First attempt within seconds of terminal state	Customers rely on timely notification, but result availability is not blocked by webhook success.
GDPR deletion workflow	Completion tracked and alerted	Compliance process must not silently stall.

5.2 Stage Metrics

Each stage has explicit latency and cost budgets that drive alerting and backpressure decisions.

Stage	Latency budget (P95)	Cost target	Key metrics
Ingestion	n/a (excluded from SLA clock)	~\$0.005 / 100 pages	jobs created, uploads completed/expired/rejected, validation latency, malware pending/rejected, page-count rejected, rate-limited requests, validation async DLQ depth.
Pre-processing	$\leq 30s$ typical	<\$0.001/100 pages	profiling latency, pages profiled, pages enhanced, page route counts, manifest write failures, unsupported page count, estimated downstream OCR pages.
OCR	$\leq 180s$ (P50 $\leq 60s$,	~\$0.01– 0.04/100	OCR queue age, OCR units active, callback latency, callback timeout

Stage	Latency budget (P95)	Cost target	Key metrics
AI extraction	P99 $\leq 240s$	pages	count, OCR page count, Textract throttles/errors, low-confidence pages, OCR cost estimate, OCR DLQ depth.
	$\leq 60s$ (P50 $\leq 20s$, P99 $\leq 90s$)	$\sim 0.017/100$ pages ($\leq \$0.025$ stop-loss)	model queue age, input/output/retrieval tokens, TTFT, model latency, throttles, schema validation failures, citation failures, repair attempts, stronger-model routes, model cost estimate.
	$\leq 30s$ (P50 $\leq 5s$, P99 $\leq 60s$)	$\leq 0.011/100$ pages ($\leq \$0.025$ hard stop-loss)	merge latency, chunks merged, field conflicts, critical conflicts, synthesis calls, synthesis tokens, degraded-review count, manual-inspection count.
Post-processing	$\leq 10s$ (P50 $\leq 2s$, P99 $\leq 20s$)	$\leq 0.001/100$ pages ($\leq \$0.005$ hard stop-loss)	validation latency, fields validated/normalized, validator failures, redacted/suppressed fields, reprocess requests, redaction failures, post-processing DLQ depth.
Storage and notification	$\leq 10s$ (P50 $\leq 2s$, P99 $\leq 20s$)	$\leq \$0.002/\text{job}$ total	output promotion latency, retrieval index failures, terminal state counts, signed URL count, webhook attempts/success/failure/retry exhaustion, deletion requested/completed.

6. Dashboards

6.1 Executive Health Dashboard

Audience: product, engineering leadership, on-call lead.

Shows:

- Jobs created, accepted, completed, failed, manual-review-required.
- End-to-end P50/P95/P99 latency.

- Job success rate.
- Cost per 100 pages.
- Manual-review and degraded-review rate.
- Provider outage/circuit-breaker status.
- Current queue backlog and age.

6.2 Pipeline Operations Dashboard

Audience: on-call engineers.

Shows:

- Stage-by-stage latency and queue age.
- DLQ and poison queue depth.
- Retry counts by stage.
- Slowest active jobs and stuck states.
- Tenant rate limiting and 429 counts.
- KMS, IAM, S3, DynamoDB, Step Functions, Textract, Bedrock errors.
- Reconciliation job findings for missed S3 events or stuck uploads.

6.3 Model Operations Dashboard

Audience: ML/platform engineers.

Shows:

- Bedrock calls by model route.
- Input/output tokens.
- TTFT and model latency.
- Throttling and error rate.
- Guardrail interventions.
- JSON/tool validation failures.
- Citation failures.
- Repair prompts and stronger-model recovery.
- Quality metrics by schema, document type, and model version.

6.4 Cost Dashboard

Audience: finance, product, platform.

Shows:

- Cost per job and per 100 pages.

- Cost by tenant, schema, document type, and stage.
- OCR cost by page route.
- Model cost by token type and route.
- Retry/replay cost.
- Storage cost by retention class.
- Scanned-heavy and structured-OCR-heavy exception rate.

6.5 Self-Hosted GPU Dashboard

Audience: ML platform and infrastructure.

Shows:

- GPU utilization and memory.
- vLLM queue depth and requests waiting.
- Tokens/sec and cost per 1M tokens.
- TTFT and model latency.
- Pod restarts and error rate.
- Karpenter node provisioning time.
- Spot interruption events.

7. Alerting Strategy

Alerts should be tied to customer impact, not every noisy metric.

Alert	Trigger	First response
End-to-end latency burn	P95 burn rate exceeds threshold	Check queue age and slowest required stage; apply admission backpressure if needed.
Stage queue age	Stage 3/4/5 queue age threatens SLA	Reduce intake, inspect provider throttles, raise service quota if sustained.
DLQ growth	Any stage DLQ grows above baseline	Identify failure reason, replay safe items, isolate poison items.
Textextract throttling/error spike	OCR throttles or errors exceed threshold	Open OCR circuit breaker, pause new OCR starts, keep accepted jobs queued.
Bedrock throttling/error	Model throttles or errors exceed	Apply model backpressure, use retry budget, route critical chunks per

Alert	Trigger	First response
spike	threshold	policy.
Cost spike	Cost per 100 pages or per tenant exceeds policy	Stop new expensive work for affected tenant/job, inspect OCR route and token usage.
Token spike	Input/output tokens exceed expected range	Inspect prompt/schema changes, few-shot retrieval, document density.
Citation failure spike	Citation validation failures increase	Roll back prompt/model/schema release if tied to canary.
Manual-review spike	Manual-review rate rises unexpectedly	Inspect OCR quality, model release, schema changes, and document mix.
PII detected in logs	Log leakage detector fires	Treat as security incident; stop offending logger/release and rotate any exposed secret if needed.
KMS/IAM failures	Access denied events rise	Pause affected tenant jobs and investigate policy/config drift.
Webhook retry exhaustion	Retry exhaustion exceeds baseline	Notify support; customer endpoint may be down or misconfigured.

8. Deployment and Release Operations

8.1 CI/CD Gates

Every change goes through GitHub Actions:

1. Format and lint.
2. Unit tests with at least 80% coverage.
3. Security scan and dependency check.
4. CDK synth.
5. IAM/policy checks.
6. Integration tests for core job lifecycle paths.
7. Model/prompt/schema evaluation when the change affects extraction behavior.
8. Deployment to a canary environment or canary traffic slice.

8.2 Canary Policy

Canaries run at 10% traffic for 1 hour.

Canary guardrails:

- No material increase in P95 latency.
- No material increase in error rate or DLQ depth.
- No material increase in cost per job or tokens per page.
- No regression in critical-field F1.
- No regression in citation validity.
- No unexpected increase in manual-review rate.
- No PII leakage events.

Rollback triggers:

- Critical-field F1 regression.
- Schema validation failure spike.
- Citation failure spike.
- Provider throttling caused by new concurrency behavior.
- Cost-per-job spike.
- Security or PII leakage event.

Rollback must support:

- Application code.
- Infrastructure changes.
- Prompt versions.
- Schema versions.
- Merge policy versions.
- Model route changes.

9. Replay and Recovery Operations

Replay is scoped. Operators should replay the smallest unit that can fix the issue.

Failure scope	Replay unit
Upload validation missed/delayed event	Stage 1 validation replay from S3 object metadata.
Pre-processing transient failure	Stage 2 replay from original input.
OCR callback missed	Poll Textract once, then replay OCR unit without duplicate paid start when possible.

Failure scope	Replay unit
OCR normalization failure	Replay normalization from completed Textract output.
Model invalid JSON	Validation/repair replay from stored prompt/model response.
Critical extraction failure	Stronger-model replay for affected chunk only.
Merge issue	Stage 5 replay from current chunk artifact hashes.
Post-processing issue	Stage 6 validation replay from Stage 5 artifact.
Webhook delivery failure	Webhook replay for same event ID and endpoint.